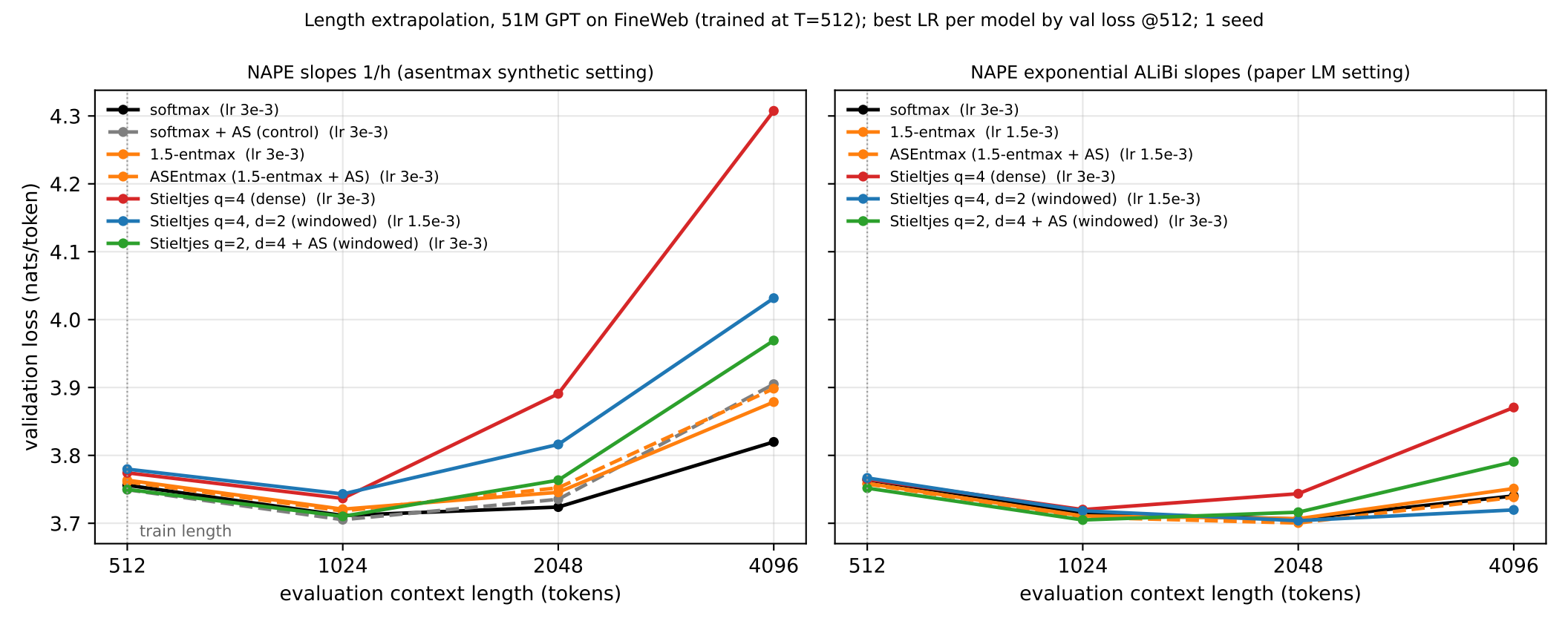
Length extrapolation, 51M GPT on FineWeb (trained at T=512); best LR per model by val loss @512; 1 seed
NAPE slopes 1/h (asentmax synthetic setting)
NAPE exponential ALiBi slopes (paper LM setting)
validation loss (nats/token)
4.3
4.2
4.1
4.0
3.9
3.8
3.7
softmax (lr 3e-3)
softmax + AS (control) (lr 3e-3)
1.5-entmax (lr 3e-3)
ASEntmax (1.5-entmax + AS) (lr 3e-3)
Stieltjes q=4 (dense) (lr 3e-3)
Stieltjes q=4, d=2 (windowed) (lr 1.5e-3)
Stieltjes q=2, d=4 + AS (windowed) (lr 3e-3)
train length
512
1024
2048
4096
evaluation context length (tokens)
softmax (lr 3e-3)
1.5-entmax (lr 1.5e-3)
ASEntmax (1.5-entmax + AS) (lr 1.5e-3)
Stieltjes q=4 (dense) (lr 3e-3)
Stieltjes q=4, d=2 (windowed) (lr 1.5e-3)
Stieltjes q=2, d=4 + AS (windowed) (lr 3e-3)
512
1024
2048
4096
evaluation context length (tokens)